Phylogeny as a proxy for structural linguistic diversity: a reasonable approach?

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In this paper, the possibility of using phylogenetic relatedness as a proxy for linguistic similarity in the context of structural linguistic diversity is explored. The cross-linguistic databases Grambank, ASJP, and PHOIBLE are assessed on their suitability for deriving such measures in the context of global linguistic diversity. After intersecting the databases, measures of linguistic similarity and tree-topology similarity are derived between all pairs of languages and correlated. The results suggest a weak to moderate association between linguistic similarity measures and tree-topology similarity, concluding that phylogeny is a limited proxy and that large-scale studies of linguistic diversity should model structural dimensions separately.

1. Introduction

The fact that the linguistic diversity that has evolved over the past millennia is more recently under threat is no news to the linguistic community: in 2003, UNESCO published its 6-level graded scale on language endangerment; Lewis and Simons (2010) expanded Fishman's GIDS to the 13-level EGIDS and Glottoscope maintains the Agglomerated Endangerment Scale (AES) (Hammarström et al., 2018) which aggregates multiple assessments of endangerment. The purpose of these scales is to measure the threat to languages becoming sleeping, i.e., not being actively used by any language communities, as opposed to being widely used and actively transmitted to the next generations. As such, this measure encodes the *stability* of languages, allowing us to model trends in language vitality—as was done by Bromham et al. (2022), who predicted a loss of 1500 languages by the end of the century. This is, however, only one aspect of diversity: the number of types, that is, languages, present, which is referred to as *richness* (Harmon & Loh, 2010). Linguistic richness—and in extension diversity—is inherently linked to language evolution, since the birth of a new language is the result of languages differentiating to the point of mutual unintelligibility. By contrast, linguistic diversity can also decrease as a result of change, as languages in contact become more similar through borrowings.

Yet, measures of linguistic diversity have not adequately accounted for the aspect of similarity, instead focusing on richness and evenness in the relative abundance of languages as measured by the proportion of L1-speakers. Harmon and

Loh's (2010) Index of Linguistic Diversity accounts for richness and evenness by measuring change in L1-speaker concentrations into dominant languages, but avoids incorporating the notion of structural—or functional—diversity: the variation in how languages function. More recently, Grin and Fürst (2022) operationalized linguistic diversity through the Hill number's framework borrowed from ecology, but shied away from incorporating the similarity of languages, discussing the problem of what aspect of language to consider: lexicon, phonology, or syntax? They suggest using phylogenetic similarity as a proxy for linguistic similarity, stating that languages that branched off later will be more similar than languages that branched off at an earlier stage. Most recently, Gurevich, Herman, Toubal, and Yotov (2025)—building on Fearon (2003) and Laitin (2000)—perpetuate this idea and use shared ancestry as a measure for *linguistic proximity*. While the idea is not unreasonable from a vertical perspective, it disregards horizontal transmission: phylogenetically unrelated languages can show meaningful similarities, and related languages can be dissimilar. While some research has quantitatively explored the relationship between phylogeny and various dimensions of linguistic similarity, such as Heeringa, Gooskens, and van Heuven (2023) and Blaschke, Fedzechkina, and Hoeve (2025), they have been limited to rather few languages. Heeringa et al. investigate lexical, phonetic, syntactical, and phylogenetic distance between 15 Germanic, Slavic, and Romance languages and generally find weak correlations between all dimensions, but a strong correlation between lexical and phylogenetic distance ($0.851 < r < 0.887$). Blaschke et al. correlate similarity along the same dimension for 263 languages from 33 language families, and report similar results, although this was not their main inquiry of interest.

In this short paper, we extend these analyses by utilizing the increase in cross-linguistic data through the Cross-Linguistic Linked Data (CLLD) project (Forkel et al., 2018). We identify the largest databases that can be used to derive measures of linguistic similarity and assess their suitability for use in measuring global linguistic diversity. We then intersect the databases and correlate linguistic similarities with similarities based on phylogeny to determine if any measure can be used as a proxy for the other, or if models of linguistic diversity should account for each dimension of linguistic function separately. Our results rather lend support for the latter.

2. Data and Methods

The databases we evaluated are: Grambank v.1.0, containing feature vectors describing grammatical function (Skirgård et al., 2023); PHOIBLE 2.0—a collection of phoneme inventories encoded in IPA (Moran & McCloy, 2019); and ASJP v.21—a collection of core vocabulary lists (Wichmann, Holman, Brown, Dryer, & Ran, 2025). We identified these as the most recent and extensive datasets for morphosyntax, phoneme inventory, and lexicon, respectively. As a reference inventory of languages to assess coverage, we used Glottolog v.5.2 (Hammarström,

Forkel, Haspelmath, & Bank, 2025). In addition, we assessed the geographic distribution of languages across macroareas from Glottolog to detect any biases toward particular regions. Since measures of diversity commonly incorporate the relative abundance of languages, we also investigated speaker coverage using a dataset synthesized by combining data from Ethnologue (Eberhard, Simons, & Fennig, 2022) and Joshua Project (2025), with speaker estimates for 6,741 languages. All analyses were conducted using R[1](R Core Team, 2024), the details of the pre-processing are outlined in the supplementary material.

We computed four different similarity measures between all language pairs in the intersection of the databases. All measures assess the extent to which the features of any two languages overlap and are restricted to values between 0 and 1, where 1 equals complete overlap. We defined **phonemic inventory similarity** between two languages as the Jaccard similarity of their phoneme inventories from PHOIBLE. This is equivalent to the strict similarity explained in further depth in Anderson et al. (2021).

Using the feature vectors from Grambank, we defined **morphosyntactic similarity** between two languages as the proportion of shared feature values out of all features for which both languages were defined, making it a similarity version of the relative Hamming distance described by Hammarström and O'Connor (2013). Since many languages are not defined for all features, we required any two languages to be defined for at least 49 common features in similarity calculations.

We operationalized **lexical similarity** using the 40-item phonemically transcribed core vocabulary lists in ASJP; for any two languages, we calculated the mean normalized Levenshtein distance (LDN) across their shared vocabulary of items using the Levenshtein algorithm (Levenshtein, 1966) as described by Petroni and Serva (2010) and subtracted it from one to yield a similarity. We applied the rule of thumb by Wichmann, Rama, and Holman (2011) to limit analysis to languages with lists containing at least 28 items.

To capture phylogenetic relatedness, we positioned ourselves close to Gurevich et al. (2025), who used Dice similarity (Dice, 1945), but instead employed Jaccard similarity due to its more intuitive interpretation.[2] As such, for any two languages, we represent each of them using a vector of Glottocodes describing the evolution of the language from the root to the node, and measure their overlap. This yields a measure based on **tree-topology similarity** (TTS), capturing coarse phylogenetic information encoded in Glottolog classifications, but—in absence of branch lengths—does not directly measure the extent of evolutionary divergence. Particularly, languages belonging to unresolved clades are modelled as equally similar, although some may have diverged further. Furthermore, since the TTS re-

[1]The code to reproduce the analysis is made available at https://github.com/Eszettfors/Phylogeny-as-a-proxy-for-structural-linguistic-diversity

[2]Notably, Jaccard similarity J can be converted to Dice similarity as per $D = 2J/(1 + J)$.

flects expert judgements, variation in treedepth is expected as a result of unequal knowledge about phylogenies across families.

Finally, we created a composite index of linguistic similarity by first normalizing all pairwise similarities describing linguistic function (morphosyntax, phoneme inventory, lexicon) to have a mean of 0 and standard deviation of 1, and then averaging across the three measures. For comparability, the index was then transformed back to the scale of 0 to 1 using min-max scaling. All pairwise similarity measures were tested for correlation using Pearson's r. We acknowledge that there exist other, more precise, ways of measuring similarity across these dimensions, e.g., all our measures are naive in the sense that they consider the difference between /d/ and /t/ the same as between /f/ and /u/, but they are straightforward to calculate and anchored in existing literature. The formal definitions of our measures are provided as supplementary material.

3. Results

The results of the coverage analysis can be seen in Figure 1. Considering the language coverage in a), ASJP has the most extensive coverage (74.44%), followed by a sharp drop to Grambank and PHOIBLE covering 29.97% and 26.53% respectively. The intersection of all three marks a further fall to 12.95%, covering only 1066 languages and thus indicating a clear lack of overlap between the databases. As such, the proportion of the languages of the world for which linguistic similarities across all dimensions can be calculated is relatively small. However, as seen in b), the coverage of speakers is across the board more extensive, with the ASJP covering close to all speakers for which we have data (97.88%). Interestingly, the speaker coverage is substantially larger in PHOIBLE (84.4%) compared to Grambank (66.65%), signaling that PHOIBLE generally covers languages with more speakers. Looking into the missing languages, the most widely spoken languages missing are from Grambank, and most prominently, these are Spanish (stan1288), with more than 450,000,000 speakers, followed by Bengali (beng1280) and Urdu (urdu1245). In contrast to the language coverage, the speaker coverage of the intersection remains substantial at 62.02%.

Furthermore, the macroarea coverage, as seen in b), shows interesting variation across the databases. Unsurprisingly, ASJP, given its extensive language coverage, follows the distribution of Glottolog well, albeit somewhat overrepresenting Africa and Papunesia while underrepresenting Eurasia. Grambank appears to be the most balanced, only underrepresenting Africa somewhat while overrepresenting Papunesia. PHOIBLE here seems to be the least balanced sample of languages, deviating from Glottolog quite clearly: both South American and Australian languages are heavily overrepresented, while languages from Papunesia constitute a comparatively small part of the sample. As a result, much of the bias of PHOIBLE is inherited by the intersection, which shows a similar but not as severe skew.

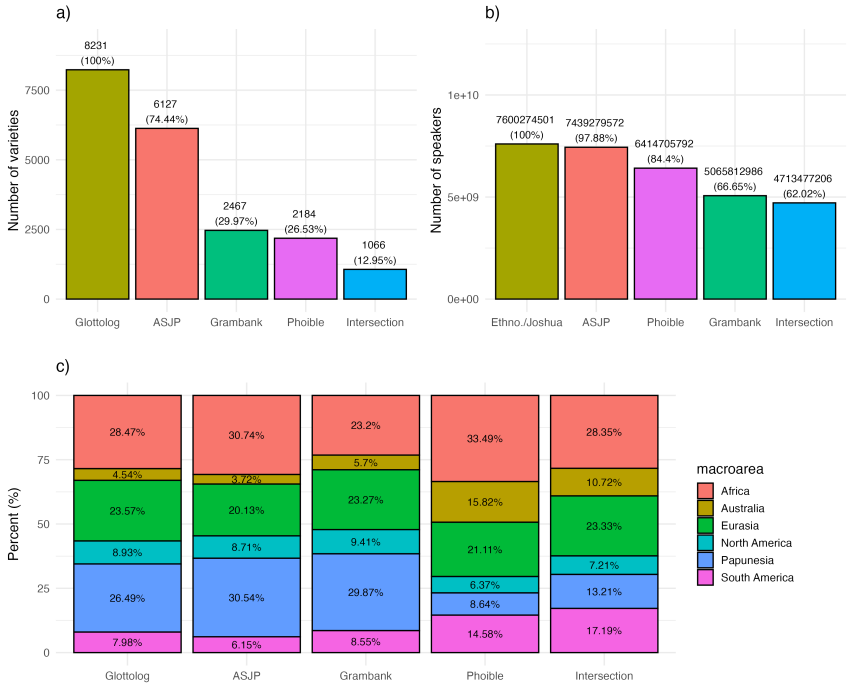

Figure 1. a) A barchart depicting the language coverage in the databases under study using Glottolog and Glottocodes as reference. b) A barchart showcasing the speaker coverage of the languages in the respective databases. c) A stacked barchart showing the distribution of languages across macroareas.

In Table 1, the correlations between the linguistic similarities can be observed[3]. Generally, no large effects are found apart from the composite index and its components. Among the components, the composite index correlates the most strongly with lexical similarity ($r = 0.695$), suggesting that lexical similarity best captures variance common to all three dimensions. Tree-topology only correlates moderately with both the composite and lexical measure ($r = 0.462; 0.457$), but apart from this, all correlations are weak. The linguistic similarities show strong independence, except for lexical similarity and phoneme inventory, which are somewhat more strongly correlated ($r = 0.246$), possibly due to ASJP using a phonemic transcription, thus capturing this signal. This association could potentially be mitigated by adding a second normalization step as described by

[3]The distribution of the pairwise similarities and scatterplots can be found in the supplementary materials at https://github.com/Eszettfors/Phylogeny-as-a-proxy-for-structural-linguistic-diversity

Wichmann, Holman, Bakker, and Brown (2010), yielding LDND instead of LDN.

Table 1. A table depicting correlations between the similarity measures as measured by Pearson's r. All correlations are significant at $p < 0.001$. n = 479,671.

	Composite index	Lexical	Morphosyntactic	Phoneme inv.	Tree-topology
Composite index	1.000	0.695	0.634	0.680	0.462
Lexical		1.000	0.150	0.246	0.457
Morphosyntactic			1.000	0.123	0.256
Phoneme inv.				1.000	0.215
Tree-topology					1.000

4. Discussion and Conclusion

We see that linguistic function is mostly independent along the dimensions of lexicon, morphosyntax, and phoneme inventory. Even though TTS is associated with the other measures—and most importantly the composite index—these correlations are too weak for it to be considered a reliable proxy. While our analysis of the correlation between the dimensions largely agrees with Blaschke et al. (2025) and Heeringa et al. (2023), identifying the strongest association as that between lexicon and phylogeny, our estimate of the effect is substantially smaller. Possibly, this is due to our analysis containing more languages and, therefore, also more pairwise similarities between languages from different families, for which TTS is 0 in our measure, even in cases where the languages show similarities. Further analyses could thus explore limiting the language pairs to those within the same macroarea or language families.

Conclusively, these results suggest that large-scale quantitative studies of linguistic diversity should model phylogenetic and structural diversity as separate from one another. However, as observed in the coverage analysis, the availability of cross-linguistic data creates a substantial bottleneck. A composite index can only be constructed for about 13% of languages, and suffers from the geographical biases of all databases. While Grambank is exemplary w.r.t. representativeness, its lack of coverage of some languages with large speaker populations makes its integration with diversity measures problematic: the diversity in the linguistic landscape of most of Latin America can't be accurately modelled without including Spanish. The ASJP, on the other hand, stands out in this regard with an extensive coverage of both languages and speakers. Furthermore, the fact that the word lists in ASJP already contain information about phonemes—which is also captured by the similarity measure—makes us identify it as the most suitable source to quantify linguistic similarity in the context of global linguistic diversity, given the current state of available data. Still, we emphasize that linguistic diversity is a multivariate phenomenon, and if the possibility of accounting for different aspects of linguistic function and structure exists, this should also be done.

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